Microsoft PM Engage 2022

Case Challenge

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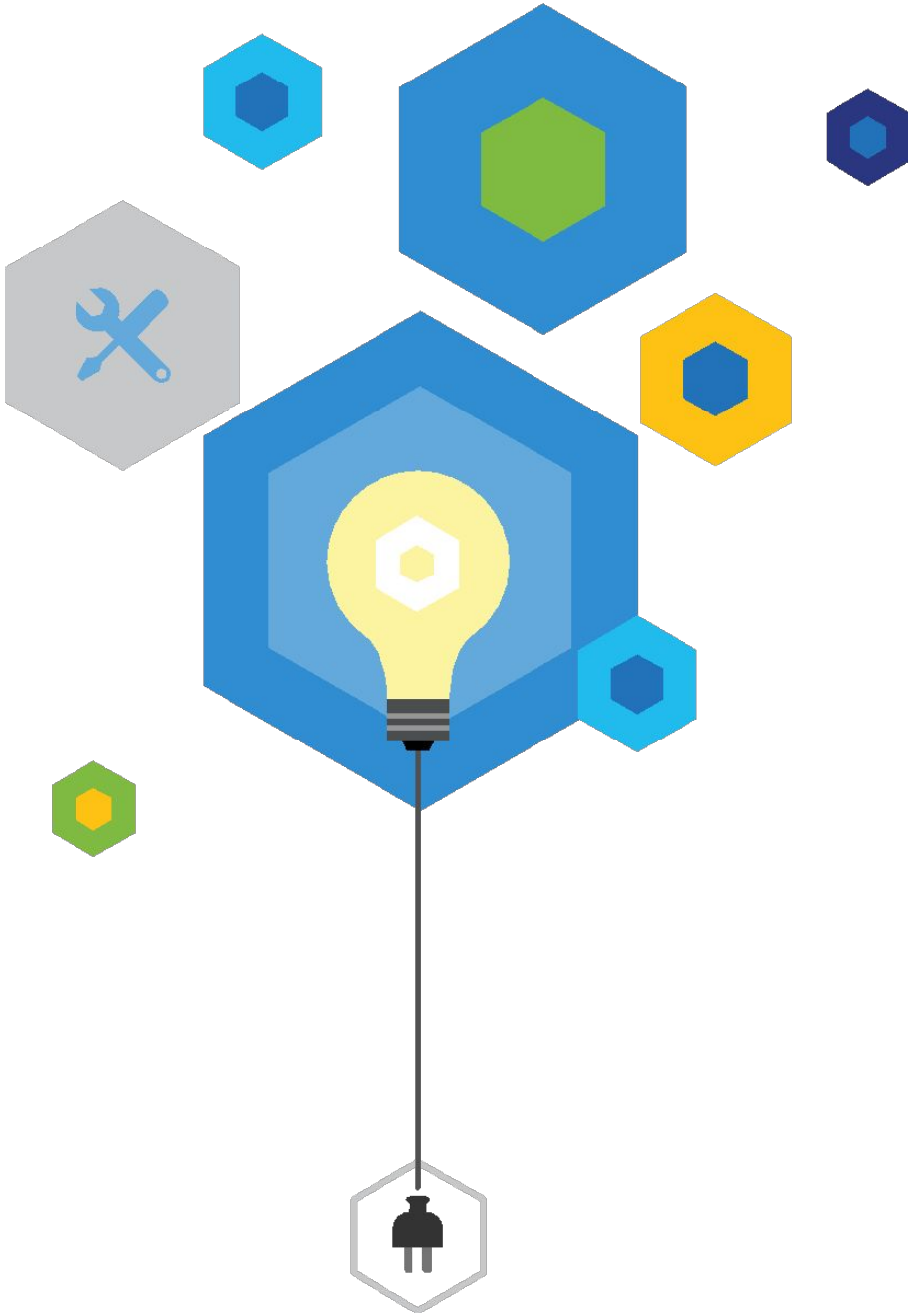
PGP/26/397

Team Gladiators

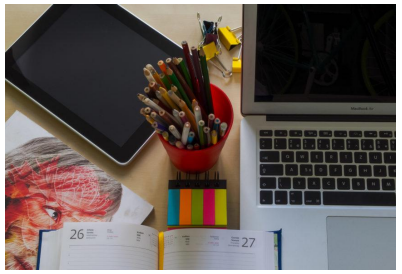
Problem Definition

Microsoft Lens is a great way to capture information from documents, whiteboards, business cards, receipts, menus, signs, handwritten memos, or anything else containing text that you want to import but not manually type out yourself.
Videos are becoming increasingly important part of data that we create and consume; for millennials(18-26 years), there is a huge opportunity to enhance user productivity by extracting rich information that exists in videos that we consume by leveraging the power of M365 ecosystem.



Personas

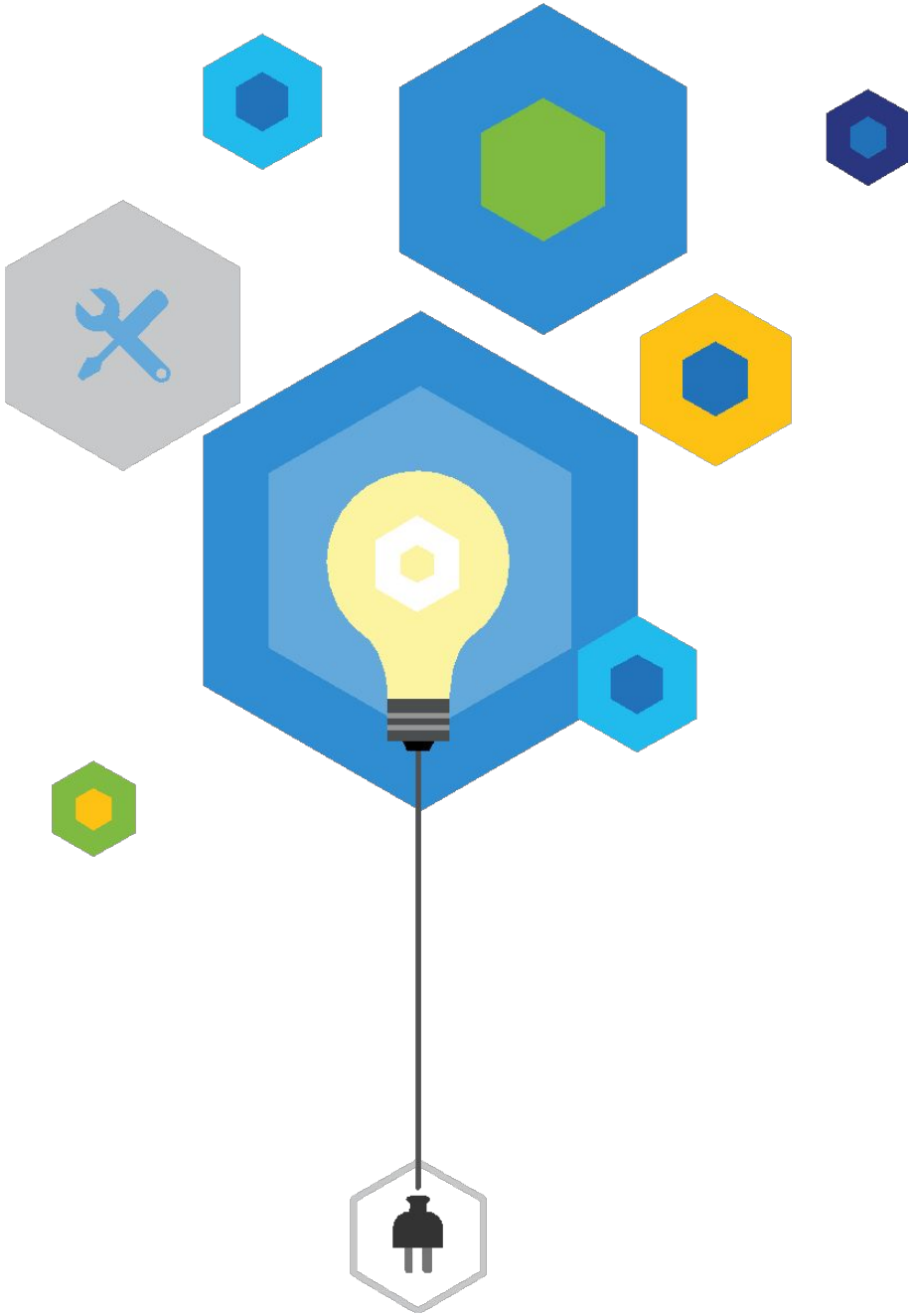
Newbie content creators - Creates high-quality content which would need advanced editing features



Students - Detail oriented , would want to cover study material

Professionals - Serious working professionals; focused on office-related work

Personal users - Casual users who would need to post life updates on social media



Customer Value

We are looking to help the millennials gain more from recording their videos

-**Productivity**- Helping users gain more from using the Microsoft Lens application by leveraging advanced video editing tools and features

-**Convenience** - Easy to understand , operate and use through seamless UI

-**Accessibility**- Increases ease of access of informative content derived from videos captured

-**Collaboration** - Gives users the features to share their videos and the content generated via the Microsoft 365 ecosystem

Business Value

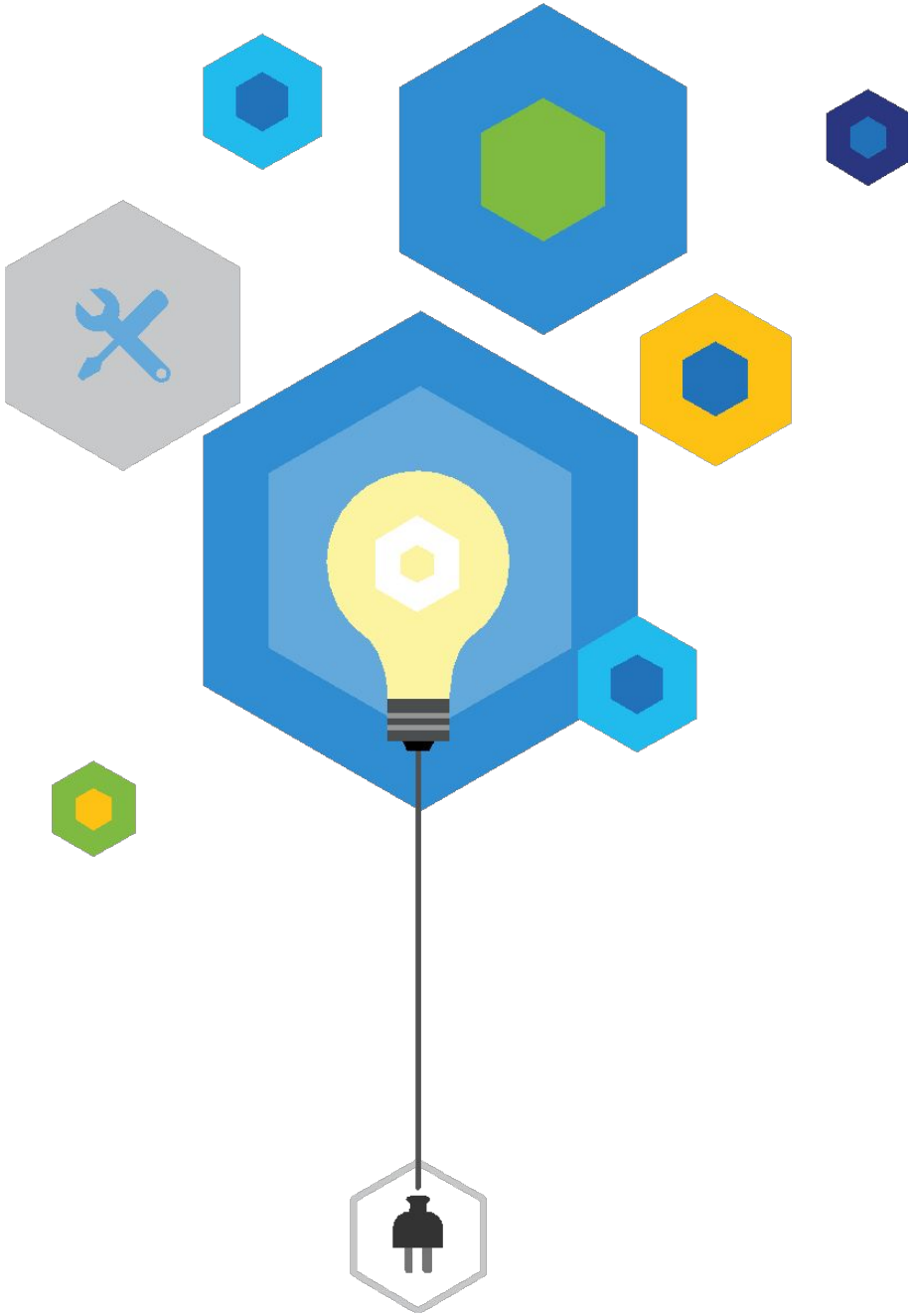
-**Customer Engagement**- Increases user engagement with multiple Microsoft 365 applications

-**User Data**- creating a database on user activity to collect better productivity tools for the future

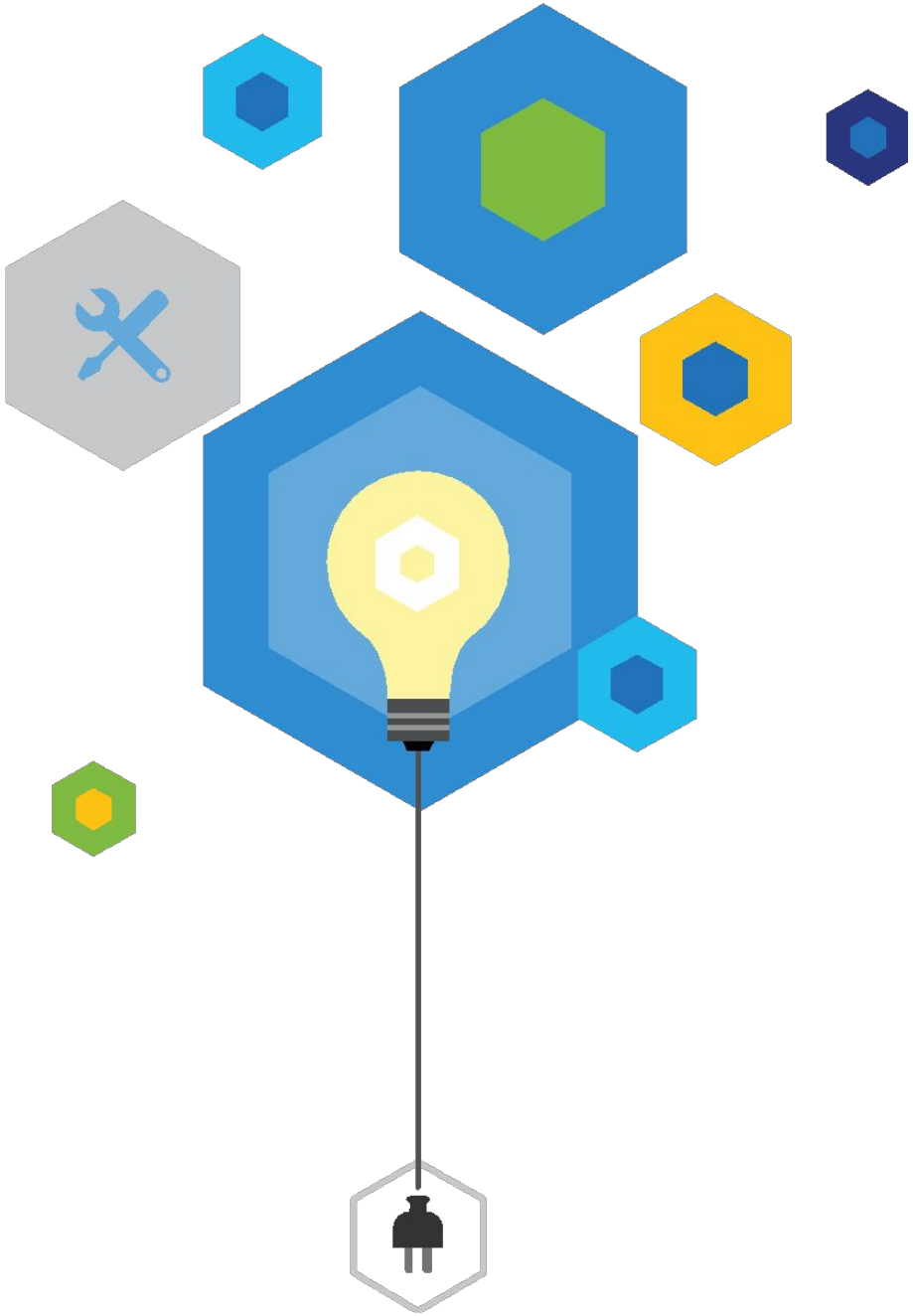
-**Customer Satisfaction** - Increase Microsoft Lens customer satisfaction to keep and win the loyalty of customers

-**Competitive Edge** -Gain a competitive edge in the current market of interactive video content creation

-**Innovation** - Innovate with advanced technologies to continuously improve the M365 ecosystem

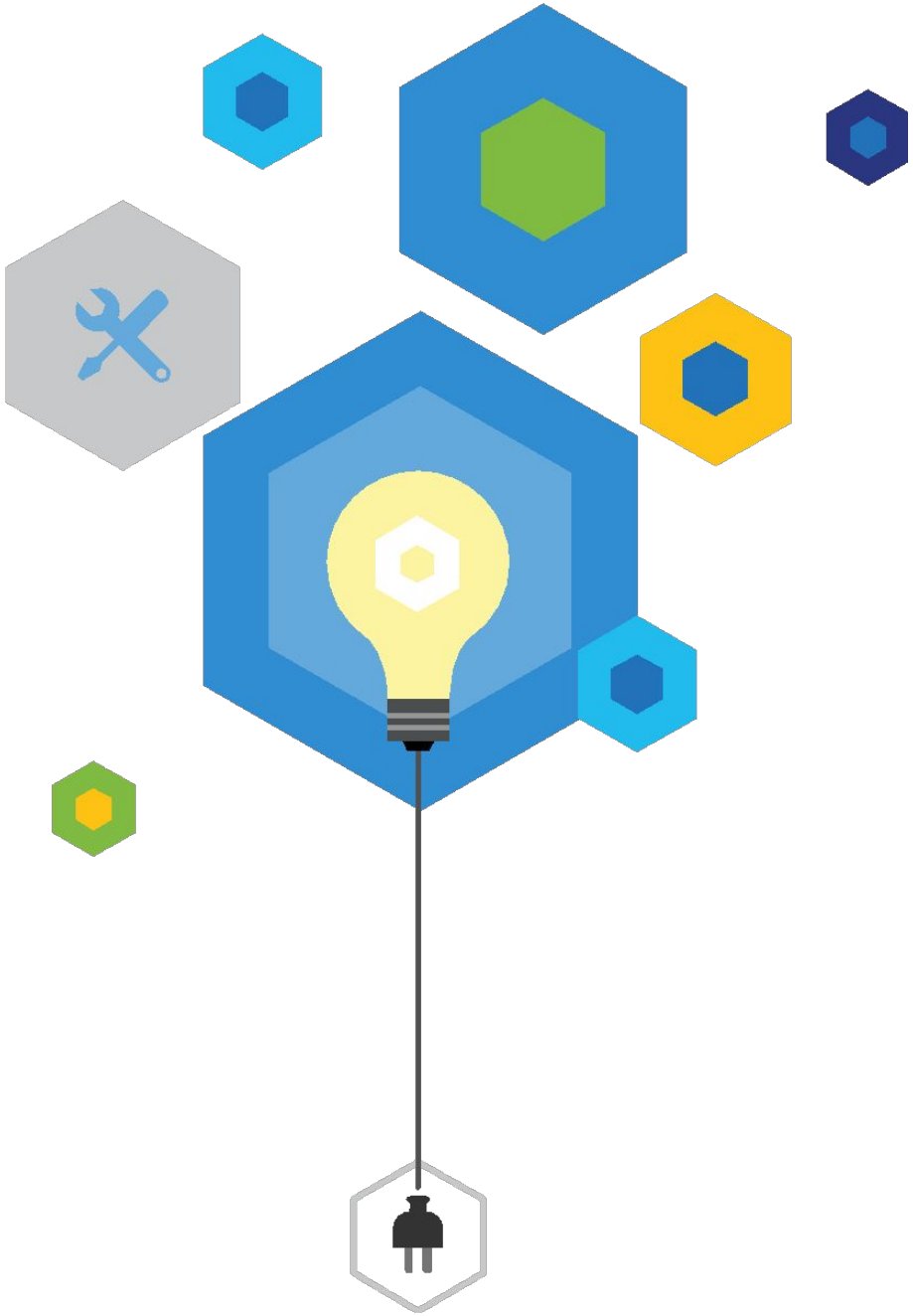


Goals	Non Goals
Easy and convenient access to enhanced video recording tool for capturing essential information	Increased user base of other M365 applications through Lens
Increased team collaboration and engagement for the tasks in hand	Collection of large video based user database for creating personalized content
Leverage advanced video editing features to make content better	
Creating an immersive platform for video content creation using the M365 ecosystem	
Increased organization of data captured through Microsoft Lens	
Increased user productivity by leveraging Microsoft Lens features	



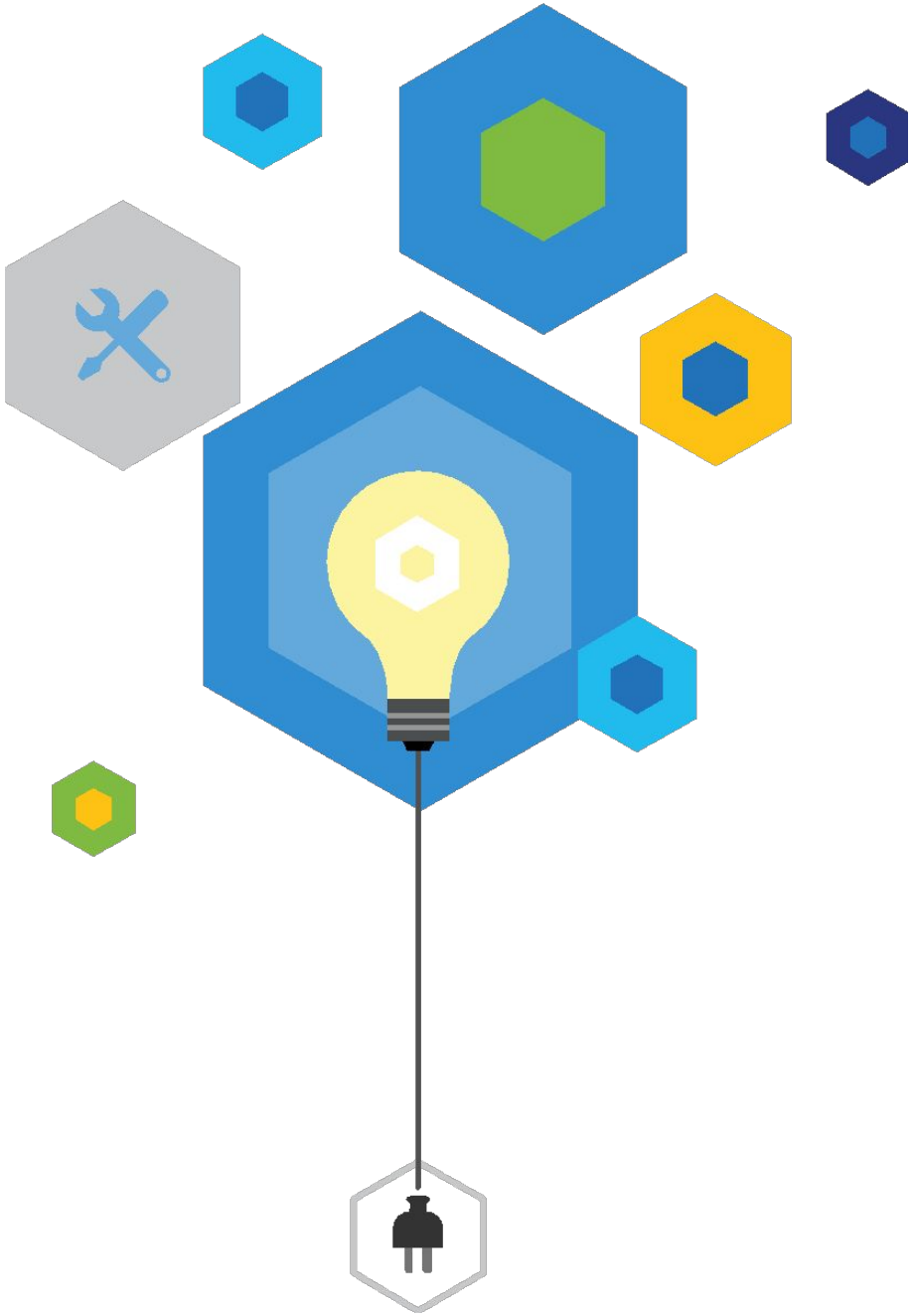
User Scenarios

Personas	Pain Points	Insights
Newbie Content Creators	Unable to create personalized video content; Loss of important content information in video while creating it; Unable to use sophisticated softwares to easily edit videos	“I want to start creating personalized video content but don't know where to start” “I am new to content creation and I don't know how to edit videos using complex softwares” “I forget the name,location,time and other details of the places I visit and videos I capture”
Students	Miss details of important lectures; Students might prefer to read information in certain instances to avoid going through videos repetitively; Finds traditional methods of teaching highly uninteractive;	“I could not recollect what my professor said during the lecture” “ I need to read the contents of the lecture” “I lose interest while listening to continuous lectures “



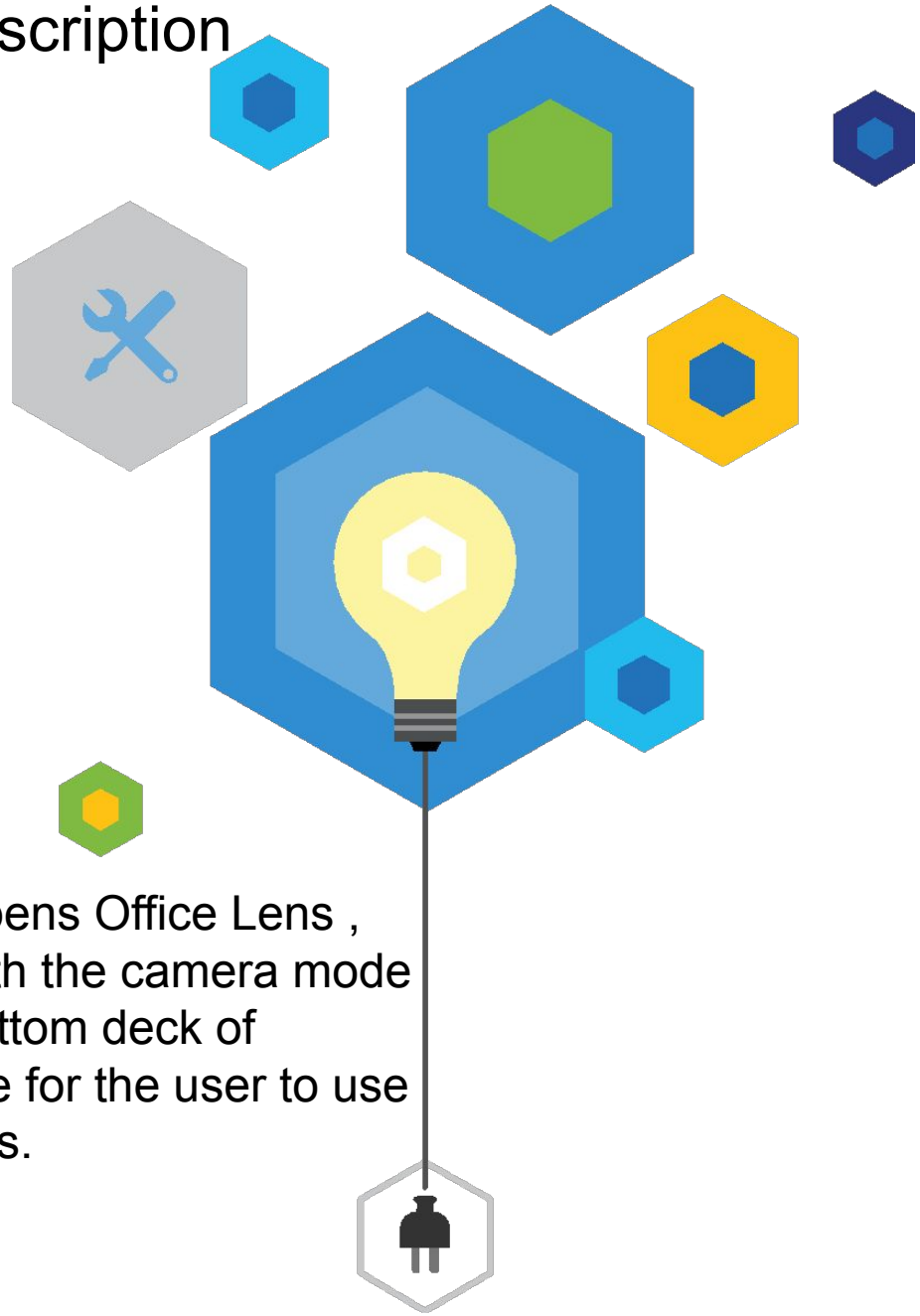
User Scenarios

Personas	Pain Points	Insights
Professionals	Lack of tools/software to share training sessions. Lack of interactive modes like slow mo which can be useful at work Lack of understanding of some visuals and audio content information which can be useful for work.	“I am a data analyst and I need to re-visit the training session conducted in my office” “I am a surgeon and I need to analyze the details of the video of my patient” “I need to record a lecture and circulate it amongst my class”
Personal Users	Need basic filters to improve the quality of video; Unable to put text/time etc into portions of video to highlight memories;	“I am unable to increase the brightness of the video I just recorded to share with my friends” “I want to use a funny face filter on my friend and share it with friends”



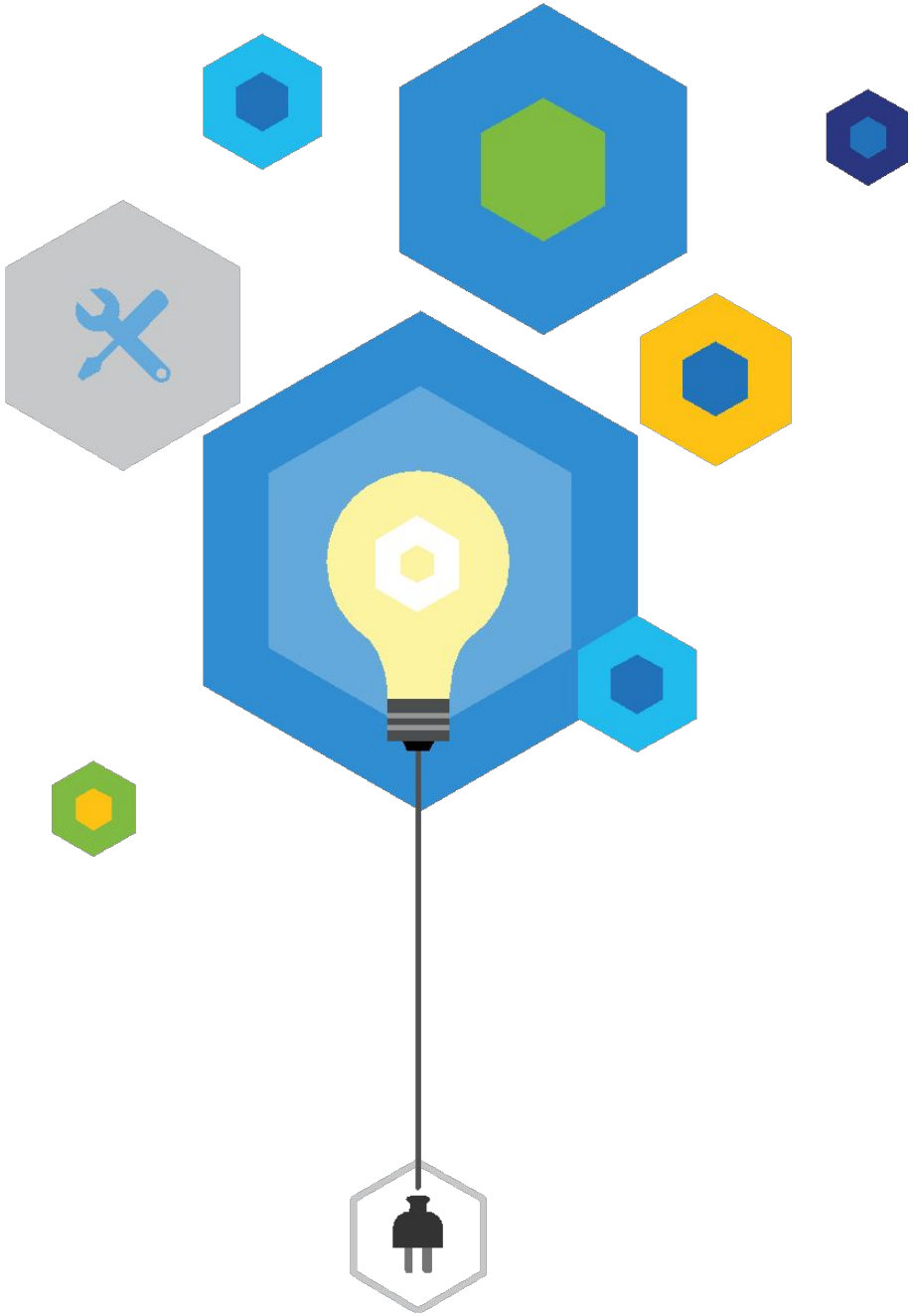
Prioritization	Feature Name	Target Persona
P0	Speech-To-Text (Record audio from videos and convert it into text)	Primary – Professionals,Students; Applicable to all
P0	Text-To-Speech (Extract Text - handwritten/printed ,from videos and convert it into speech)	Primary – Professionals,Students; Applicable to all
P0	Record audio from video and translate it into different languages	Applicable to all
P0	Understand Type of Text Detected using Natural Language Processing (Date/Email/Contact)	Applicable to all
P0	Video/Image Search	Applicable to all
P1	Audio/Voice Modulation	Primary-Professionals,Student Applicable to all
P2	Extraction of Contextual Information(Time/ Place etc)	Primary-Newbie Content Creators,Personal Users; Applicable to all
P2	Face Detection (Filters)	Primary-Newbie Content Creators,Personal Users
P3	Video Editing (Slow-motion,Fast Forward,Auto-Timer,Filters,TimeLapse,Panaroma,Snippets,Insert Text/Images/Audio)	Primary-Newbie Content Creators,Personal Users

Feature Description

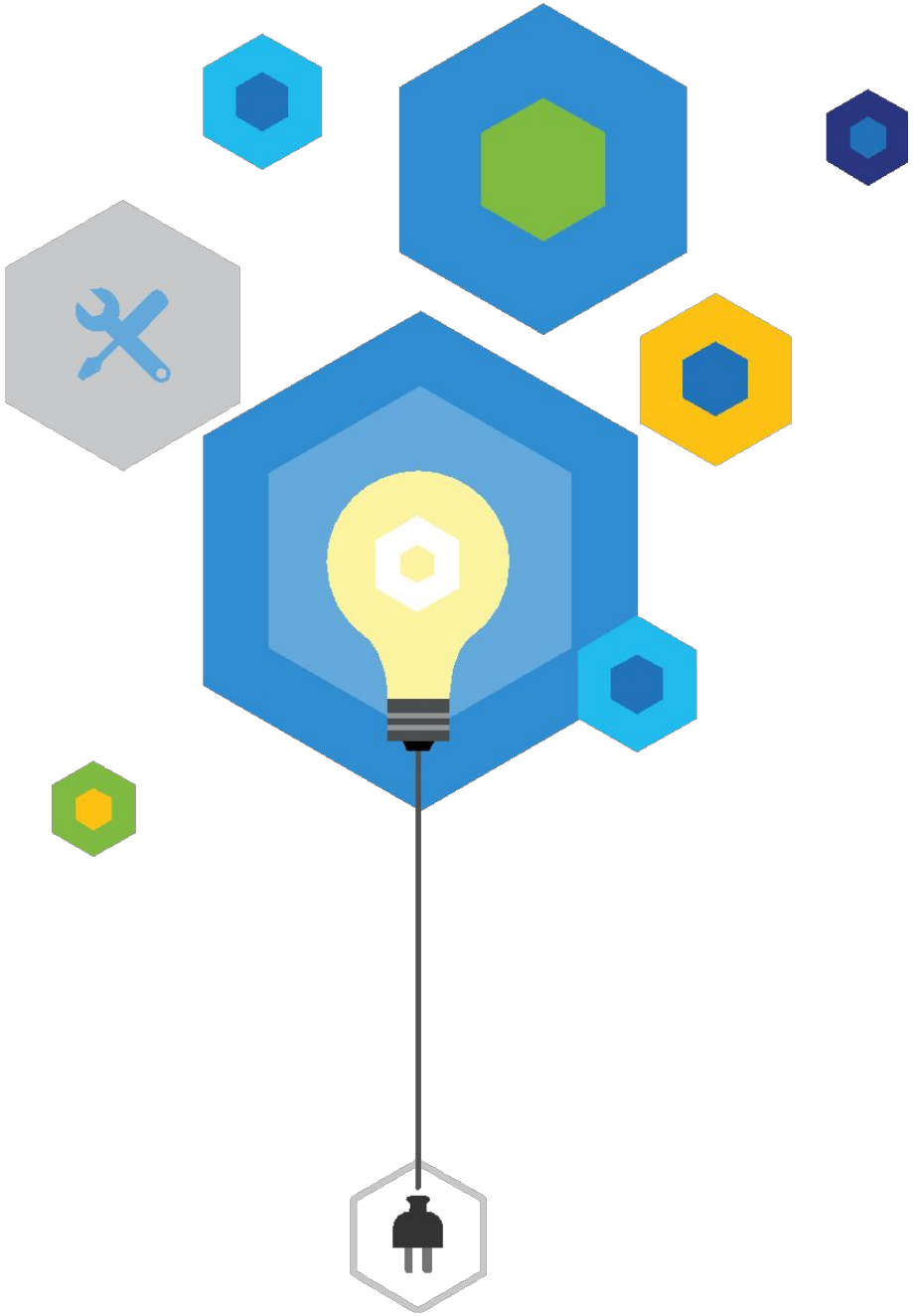


Once the user opens Office Lens , user is shown with the camera mode on .There is a bottom deck of features available for the user to use as per their needs.

- **<u>Speech-To-Text</u>**
  Feature description -
  This feature will help the user to extract text from the audio of the video recorded. This feature would be highly useful for people who are recording a video which they want to revisit later , for example class lecture,office meeting etc.
  Implementation -
  Microsoft Speech to Text API integration with Office Lens app to convert audio to text
  <u>User Interaction-</u>
  Step 1: Open Microsoft Lens application and record video
  Step 2: Click on Video Attributes tab to open the video attributes page
  Step 3: Click on Audio to Text button
  Step 4: Tap to save the result on OneDrive/OneNote/Teams

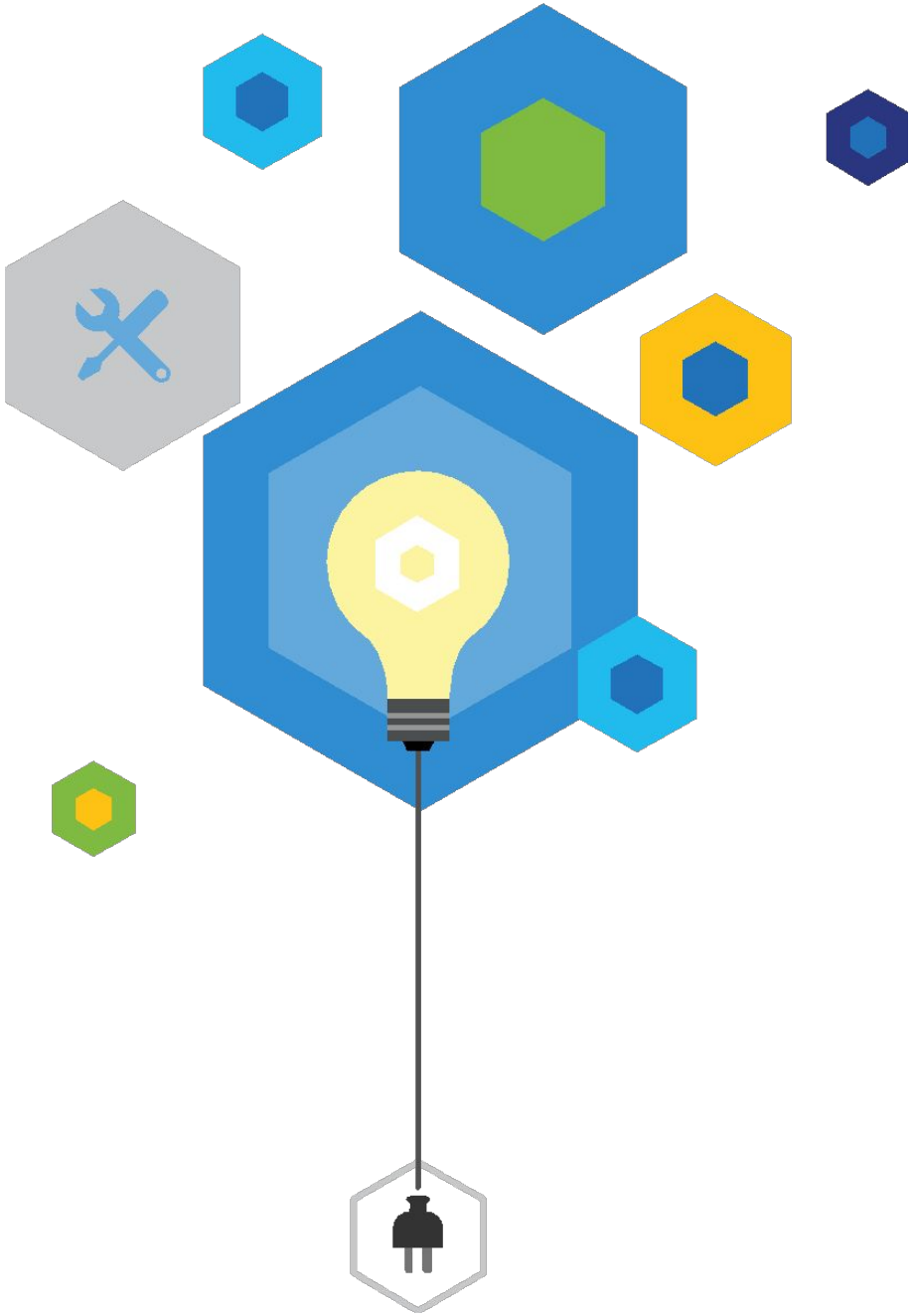


- **<u>Text-To-Speech</u>**
  Feature description -
  This feature would first extract text in different forms(handwritten/printed/typed) and then provide the user with an option to convert it into audio.
  Implementation -
  Computer Vision Read API integration with Office Lens app to extract text from the video and then using the Text to Speech API
  <u>User Interaction-</u>
  Step 1: Open Microsoft Lens application and record video
  Step 2: Click on Video Attributes tab to open the video attributes page
  Step 3: Click on Text in Video button
  Step 4: In the Text Extracted tab , tap on “Listen” button
  Step 5:Save/Share recording on OneDrive/OneNote/Teams



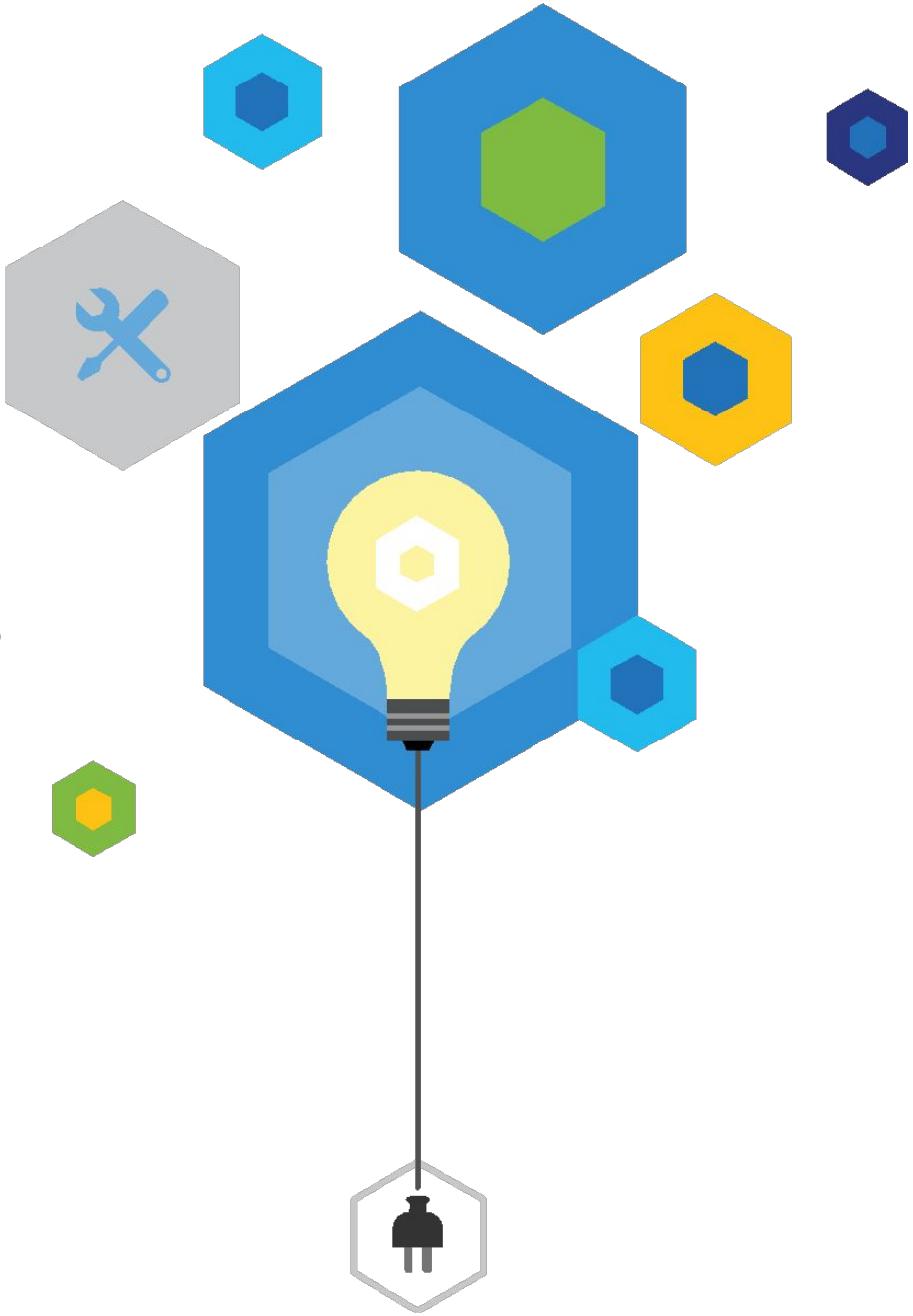
- **Translation**

  Feature description -
  This feature would first extract text in different forms(handwritten/printed/typed) and then provide the user with an option to convert it into different languages
  Implementation -
  Computer Vision Read API integration with Office Lens app to extract text from the video and then using the Text to Speech API
  <u>User Interaction-</u>
  Step 1: Open Microsoft Lens application and record video
  Step 2: Click on Video Attributes tab to open the video attributes page
  Step 3: Click on Text in Video button
  Step 4: In the Text Extracted tab , tap on “Translate” button
  Step 5:Choose Language
  Step 6:Save/Share converted text on OneDrive/OneNote/Teams

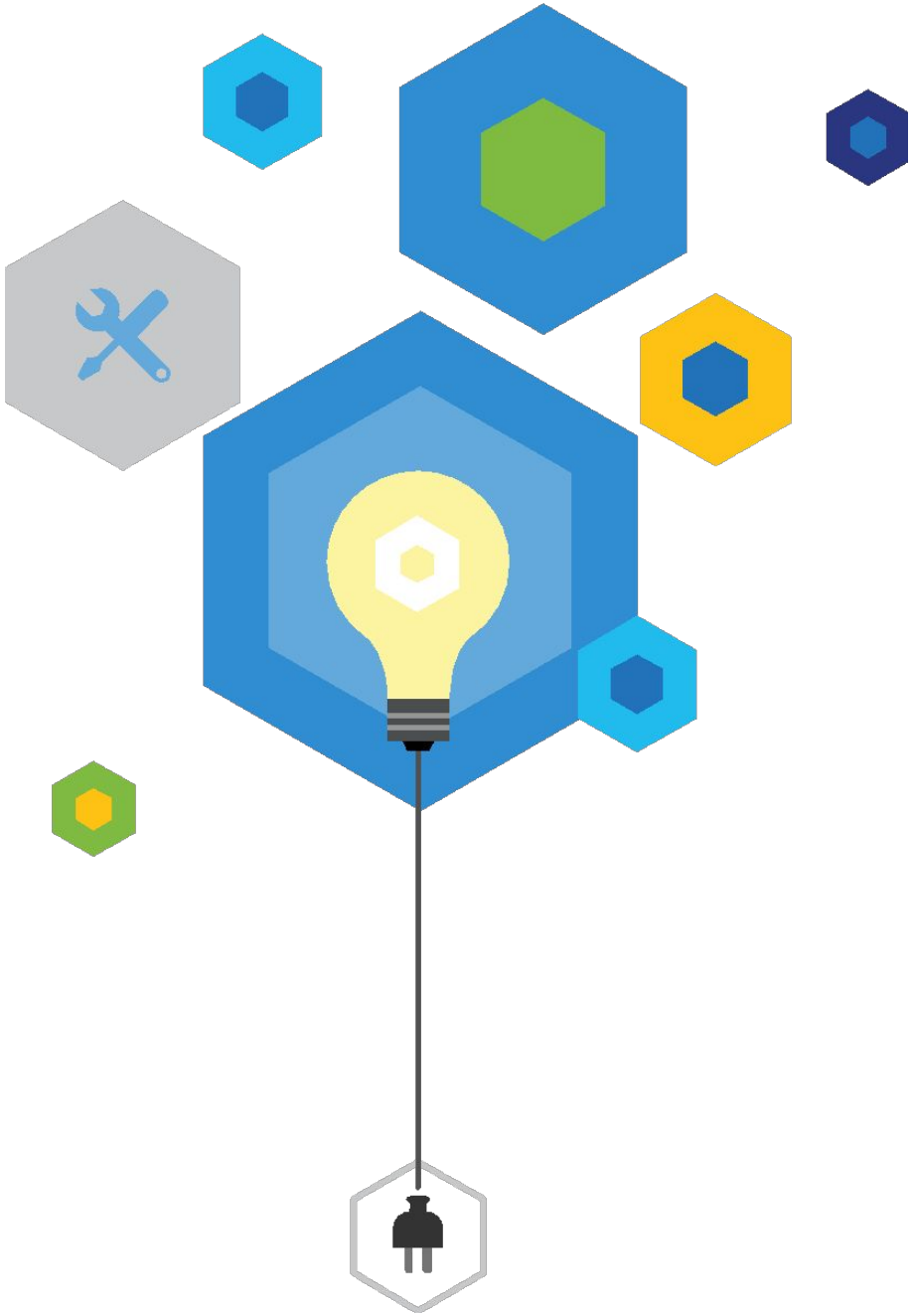


- **Understand Type of Text Detected using Natural Language Processing**

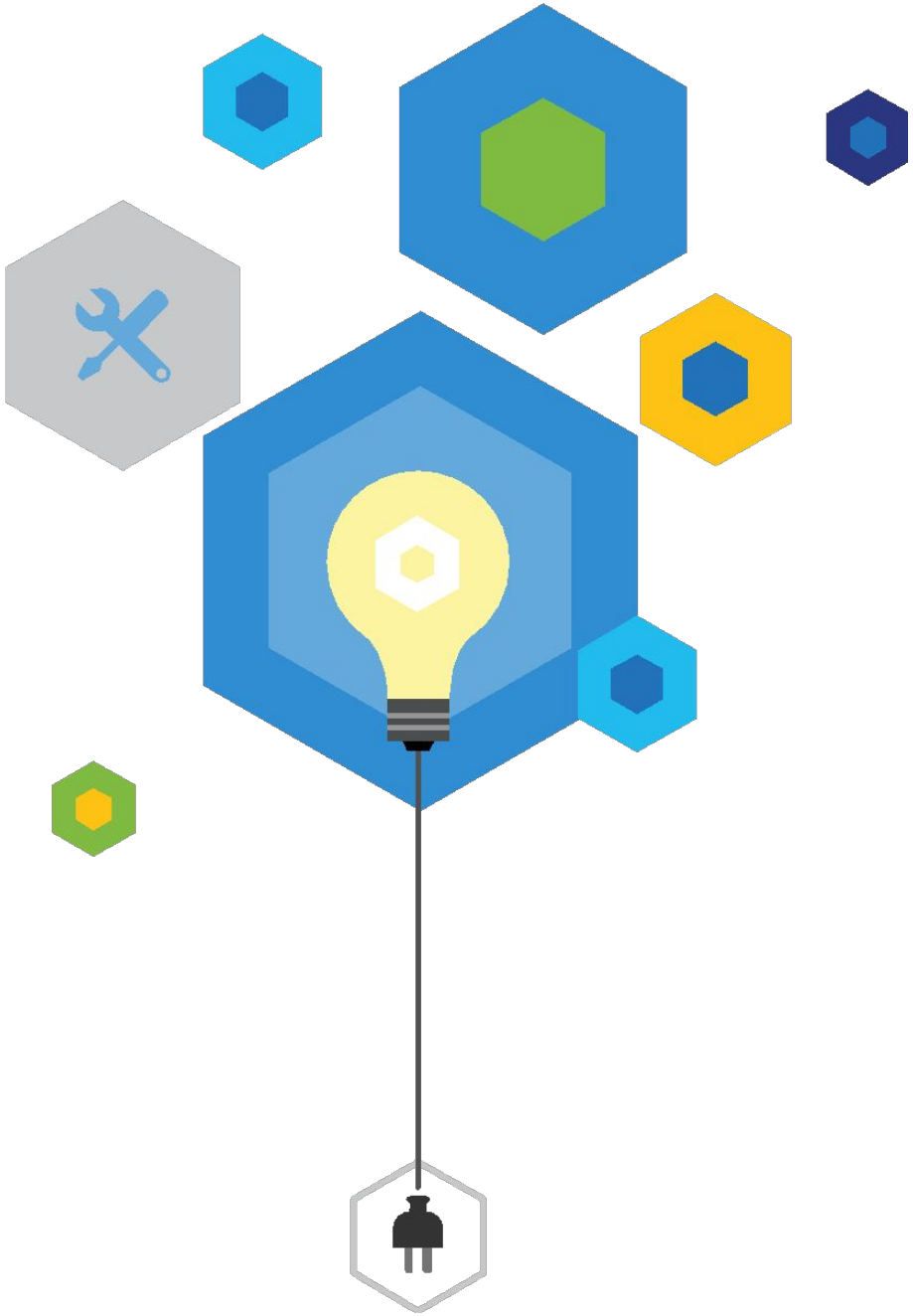
  Feature Description -
  This feature would first extract text in different forms(handwritten/printed/typed) and then auto-detect the type of text in it (date/contact/email)
  Implementation -
  Computer Vision Read API integration with Office Lens application to extract text from the video and then auto-detect the type of text extracted using NLP
  <u>User Interaction-</u>
  Step 1: Open Microsoft Lens application and record video
  Step 2: Click on Video Attributes tab to open the video attributes page
  Step 3: Click on Text in Video button
  Step 4: In the Text Extracted tab , tap on “Translate” button
  Step 5:Choose Language
  Step 6:Save/Share converted text on OneDrive/OneNote/Teams



- **Video/Image Search**
  Feature Description -
  This feature would help the user search for the video / image in the video using the Internet
  Implementation -
  Microsoft Bing API integration with Office Lens application to search for image/video
  <u>User Interaction-</u>
  Step 1: Open Microsoft Lens application and record video
  Step 2: Click on Search button
  Step 3: Select video/image
  Step 4: View the results



- **Audio/Voice Modulation**
  Feature Description -
  This feature would help the user modulate the audio recorded to amplify/reduce it accordingly
  Implementation -
  Voice Modulation API integration with Office Lens application
  <u>User Interaction-</u>
  Step 1: Open Microsoft Lens application and record video
  Step 2: Click on Video Attributes tab to open the video attributes page
  Step 3: Click on Voice Modulation button
  Step 4: Click on amplify/reduce
  Step 4: Save/Share audio on OneDrive/OneNote/Teams



- **<u>Extraction of Contextual Information</u>**

  Feature Description -

  This feature would help the user extract contextual information like time/geography/temperature etc of the video

  Implementation -

  Microsoft Bing API integration with Office Lens app

  <u>User Interaction-</u>

  Step 1: Open Microsoft Lens application and record video

  Step 2: Click on information to extract(time/geography/temperature)

  Step 3: Save/Share on OneDrive/OneNote/Teams

- **<u>Face Detection (Filters)</u>**

Feature Description -This feature would auto-detect faces in videos and help the user recognize faces and apply filters

Implementation -

Microsoft Face API integration with Office Lens application

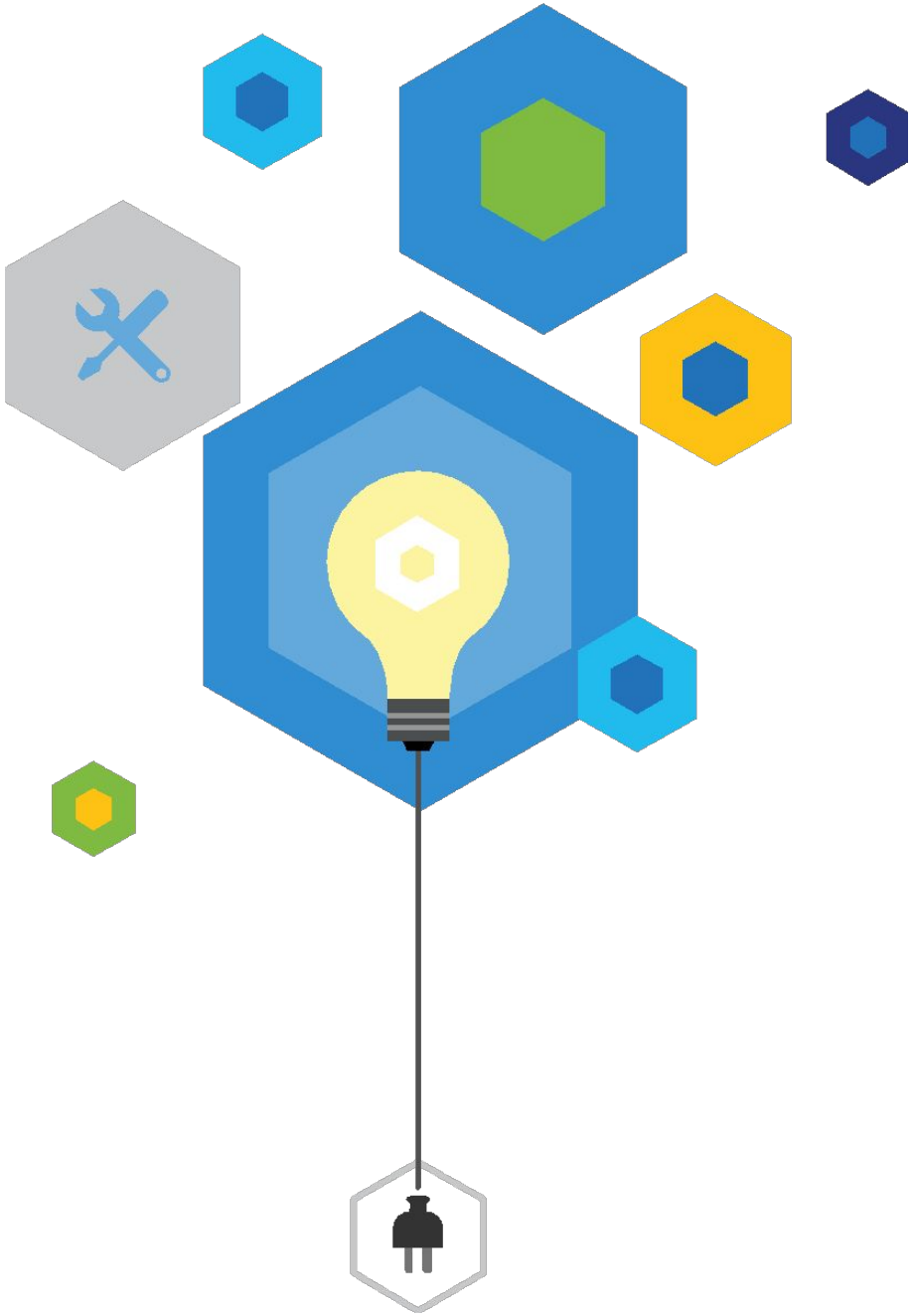
<u>User Interaction-</u>

Step 1: Open Microsoft Lens application and record video

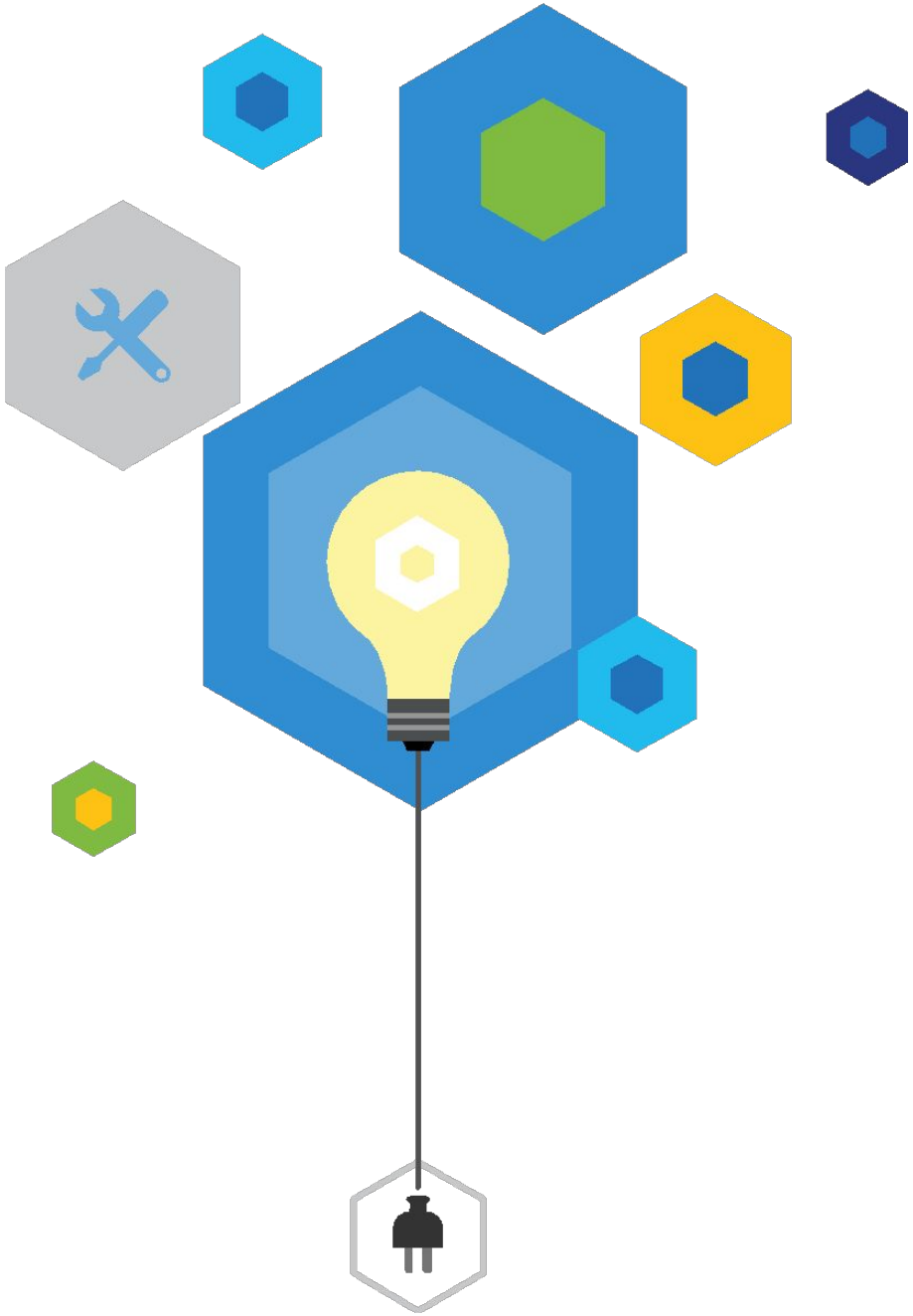
Step 2: Click on faces

Step 3: Apply filters/save information of people

Step 4:Save/Share on OneDrive/OneNote/Teams



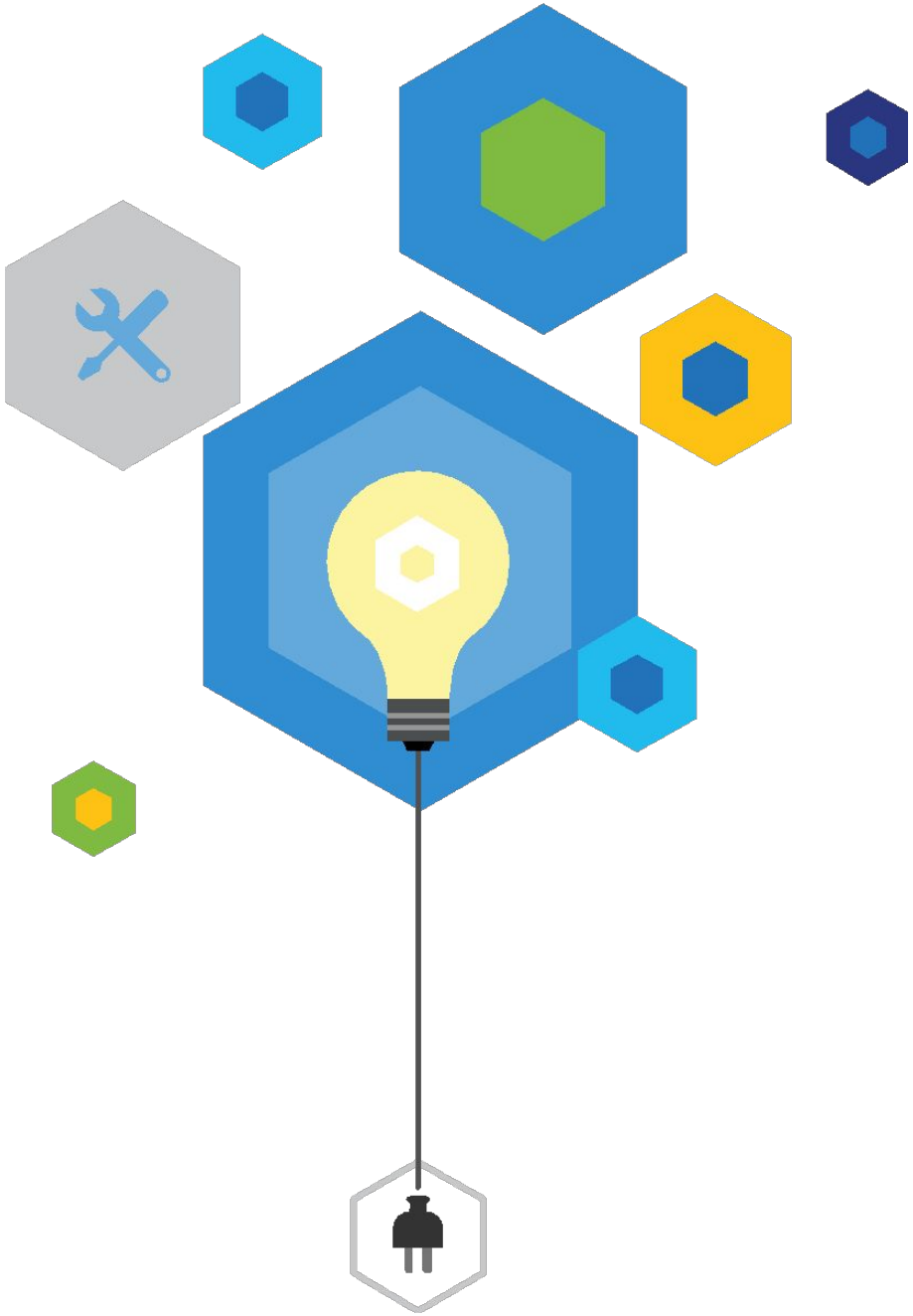
- **Video Editing**
  This feature would help user edit the video recorded and would contain a wide array of options like Slow- motion, Fast Forward, Filters, Time Lapse, Panaroma, Snippets, Insert Text/Images/Audio
  Implementation
  Advanced video editing APIs with Office Lens application
  <u>User Interaction-</u>
  Step 1: Open Microsoft Lens application and record video
  Step 2: Click on edit button
  Step 3: Apply Slow Motion, Fast Forward, Image Filters, TimeLapse,Panaroma,Snippets and Insert Text/Images/Audio
  Step 4:Save/Share on OneDrive/OneNote/Teams



Success Metrics

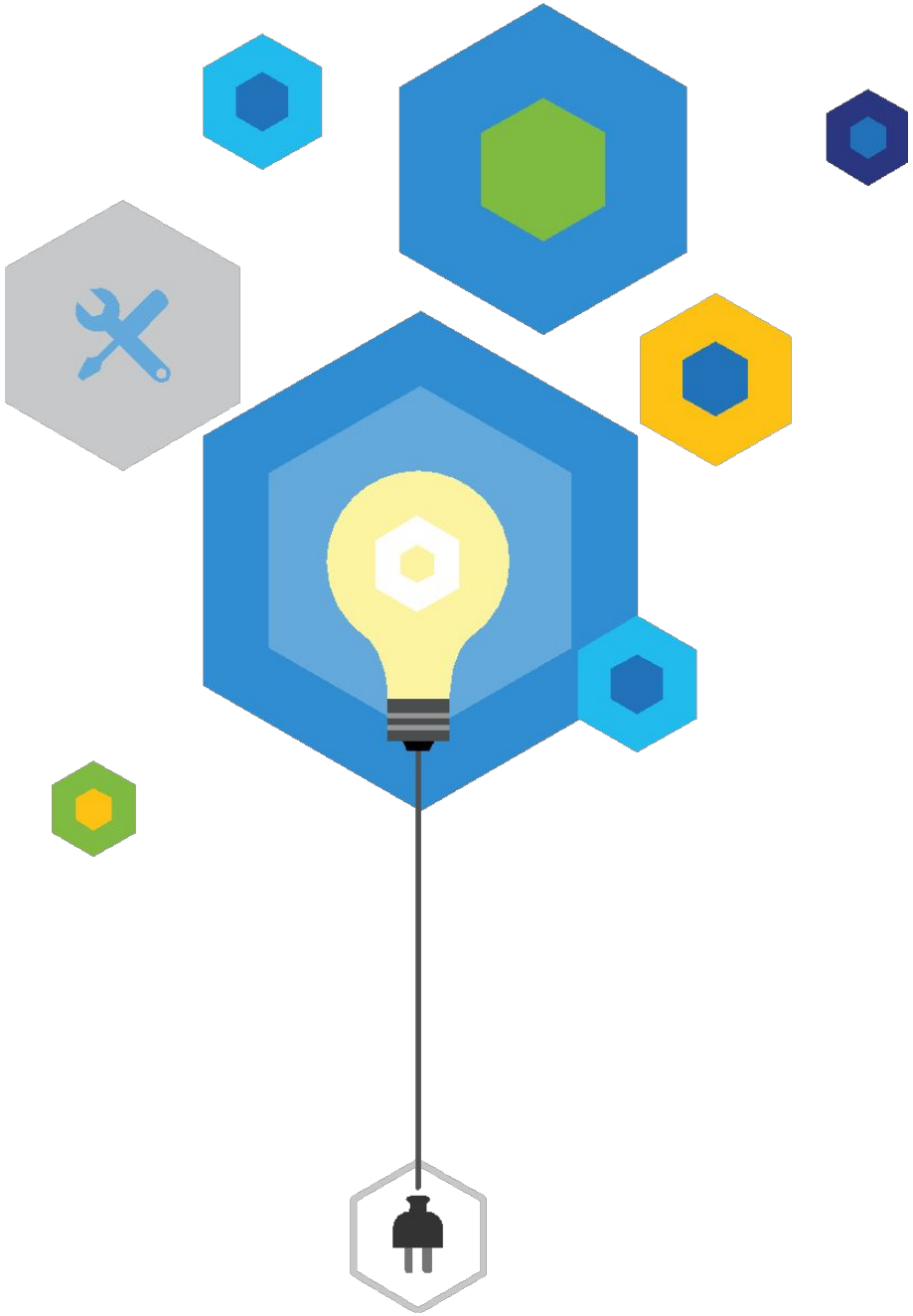
Feature metrics

Metric	Significance
Number of videos captured using the advanced tools	To understand how frequently the video recording is used per user
Number of times features in Advanced Tool used	To understand customer usefulness and understanding of the feature
Total time spent interacting with the features in Advanced tools	To understand customer convenience while using the features
Number of redirects to Office 365 apps while using the features	To understand usefulness of attributes generated to user using the Office Apps
Number of times videos shared using office	Indicates the extent and ease of collaboration of the video and attributes captured.

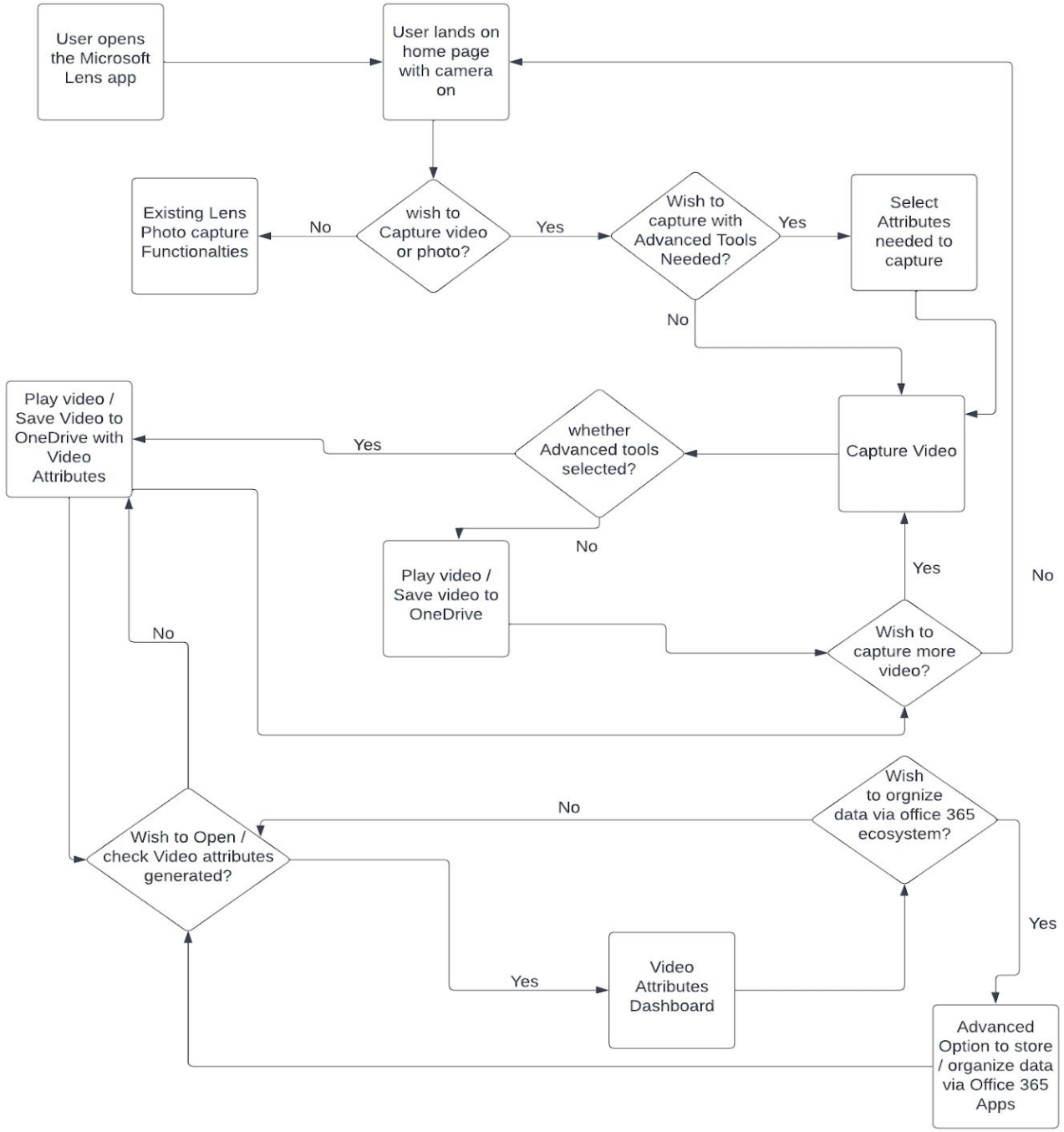


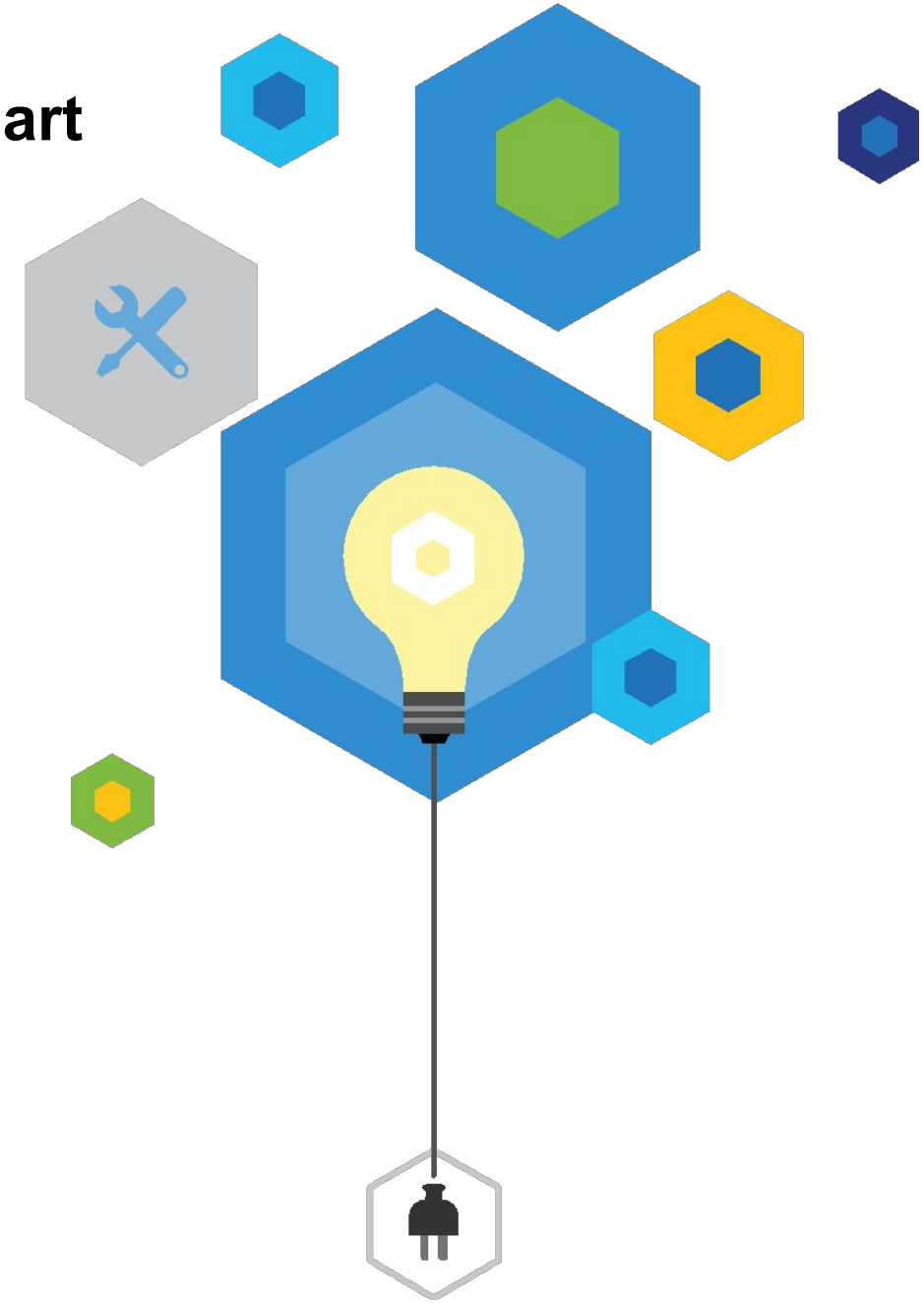
Business metrics

Metric	Significance
New Users of the Microsoft Lens application	Shows the new users started using Microsoft Lens after addition of new features
Change in user statistics for Office 365 ecosystem's application	Indicates user engagement with office 365 application due to addition of new features
Daily active users Indicates the usage of platform on a daily basis	Indicates the daily active user of lens application
Weekly active users Indicates the monthly persistent usage of platform	Indicates the monthly active user with persistent usage of the Lens application
Change in statistics of cloud storage used	Indicates the extent of usage of the Lens application

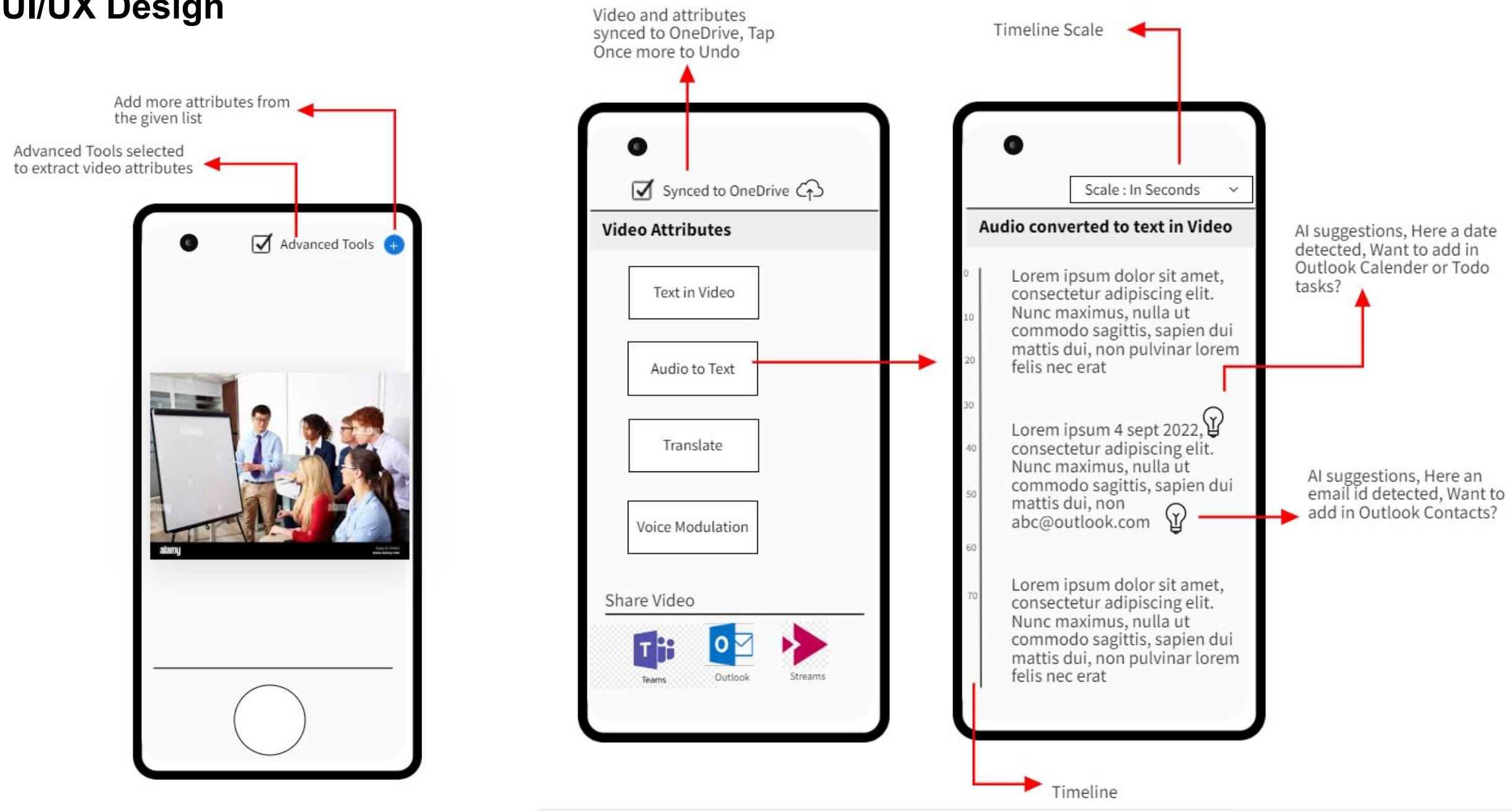


Flowchart


User opens the Microsoft Lens app
User lands on home page with camera on
wish to Capture video or photo?
No
Existing Lens Photo capture Functionalties
Yes
Wish to capture with Advanced Tools Needed?
Yes
Select Attributes needed to capture
No
Capture Video
whether Advanced tools selected?
Yes
Play video / Save Video to OneDrive with Video Attributes
No
Play video / Save video to OneDrive
Yes
Wish to capture more video?
No
No
Wish to Open / check Video attributes generated?
Wish to orgnize data via office 365 ecosystem?
No
Yes
Video Attributes Dashboard
Yes
Advanced Option to store / organize data via Office 365 Apps



UI/UX Design


Add more attributes from the given list
Advanced Tools selected to extract video attributes
Advanced Tools
Video and attributes synced to OneDrive, Tap Once more to Undo
Synced to OneDrive
Video Attributes
Text in Video
Audio to Text
Translate
Voice Modulation
Share Video
Teams
Outlook
Streams
Timeline Scale
Scale : In Seconds
Audio converted to text in Video
0
10
20
30
40
50
60
70
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AI suggestions, Here a date detected, Want to add in Outlook Calender or Todo tasks?
AI suggestions, Here an email id detected, Want to add in Outlook Contacts?
Timeline